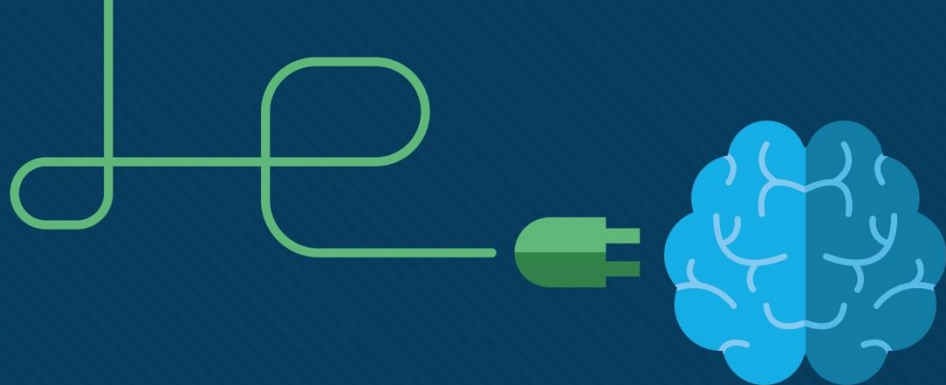




IP Addressing

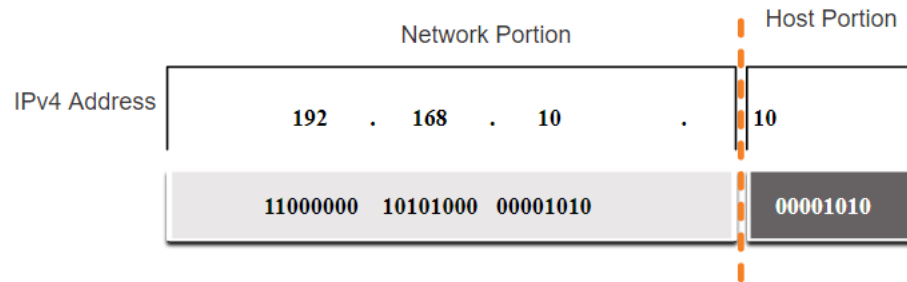


IPv4 Address Structure

IPv4 Address Structure

Network and Host Portions

- An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.
- When determining the network portion versus the host portion, you must look at the 32-bit stream.
- A subnet mask is used to determine the network and host portions.



IPv4 Address Structure

The Subnet Mask

- To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.
- The actual process used to identify the network and host portions is called ANDing.

	Network Portion				Host Portion
IPv4 Address	192	.	168	.	10
	11000000	10101000	00001010		00001010
Subnet Mask	255	.	255	.	0
	11111111	11111111	11111111		00000000

IPv4 Address Structure

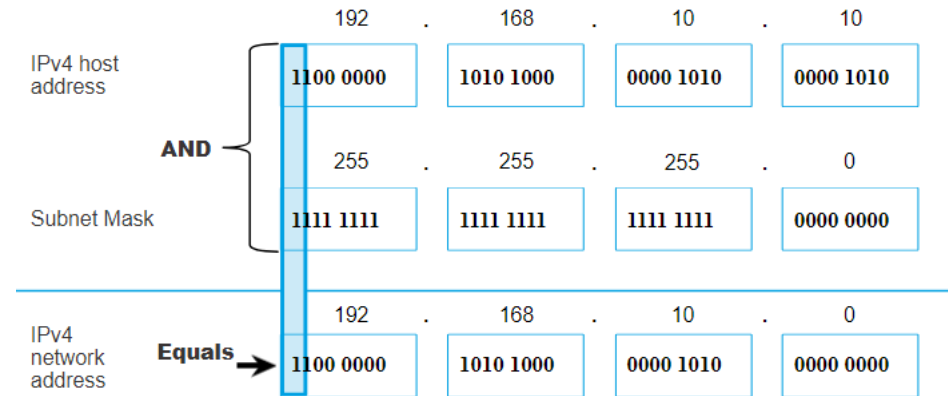
The Prefix Length

- A prefix length is a less cumbersome method used to identify a subnet mask address.
- The prefix length is the number of bits set to 1 in the subnet mask.
- It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

Determining the Network: Logical AND

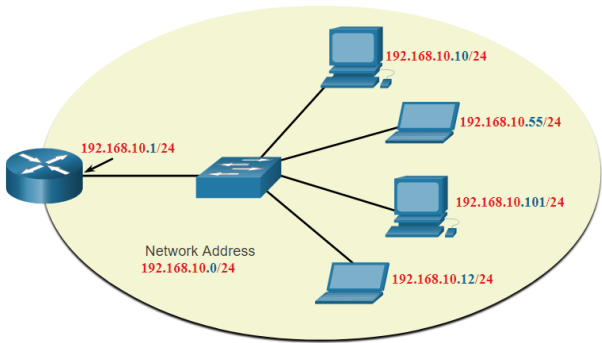
- A logical AND Boolean operation is used in determining the network address.
- Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
- $1 \text{ AND } 1 = 1$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 0 = 0$, $0 \text{ AND } 0 = 0$
- $1 = \text{True}$ and $0 = \text{False}$
- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.



IPv4 Address Structure

Network, Host, and Broadcast Addresses

- Within each network are three types of IP addresses:
 - Network address
 - Host addresses
 - Broadcast address



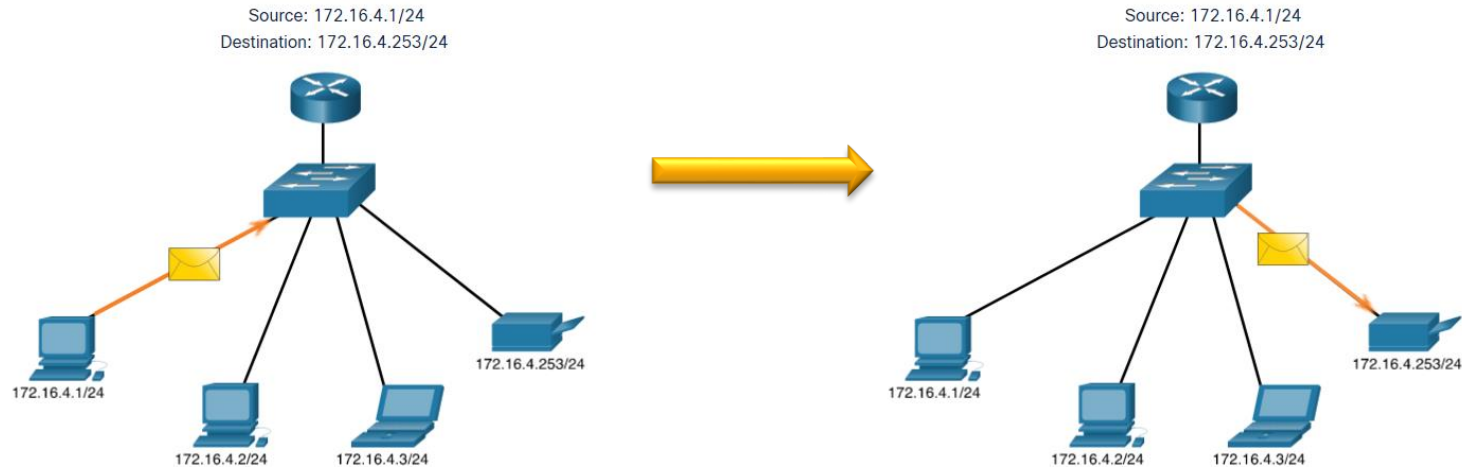
	Network Portion			Host Portion	Host Bits
Subnet mask 255.255.255.0 or /24	255 11111111	255 11111111	255 11111111	0 00000000	
Network address 192.168.10.0 or /24	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
First address 192.168.10.1 or /24	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
Last address 192.168.10.254 or /24	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
Broadcast address 192.168.10.255 or /24	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

IPv4 Unicast, Broadcast, and Multicast

IPv4 Unicast, Broadcast, and Multicast

Unicast

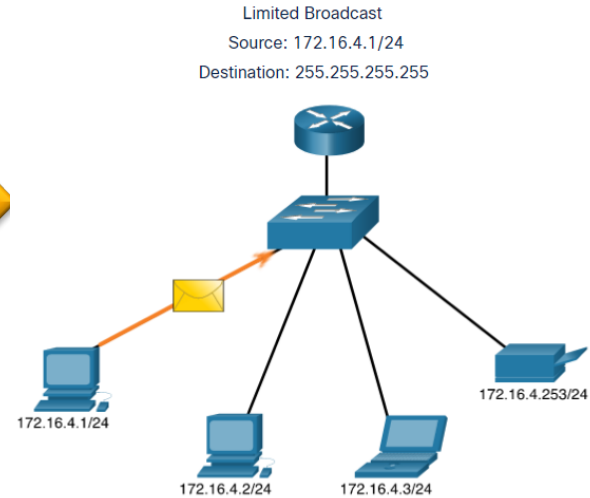
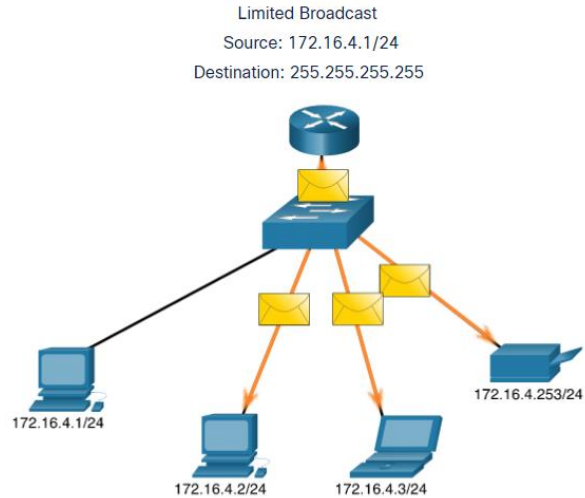
- Unicast transmission is sending a packet to one destination IP address.
- For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253.



IPv4 Unicast, Broadcast, and Multicast

Broadcast

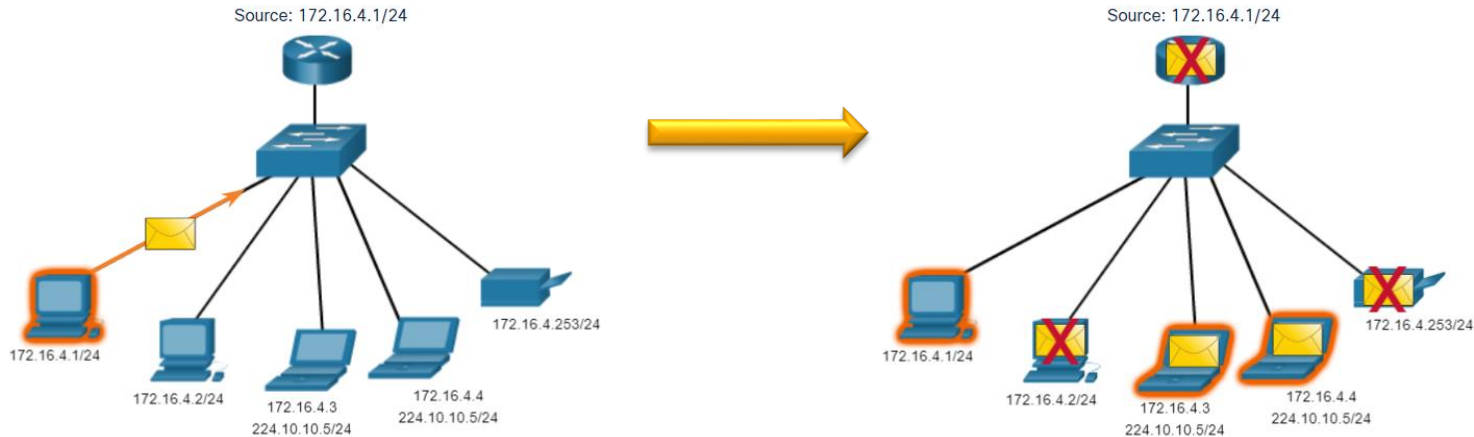
- Broadcast transmission is sending a packet to all other destination IP addresses.
- For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.



IPv4 Unicast, Broadcast, and Multicast

Multicast

- Multicast transmission is sending a packet to a multicast address group.
- For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.



Types of IPv4 Addresses

Public and Private IPv4 Addresses

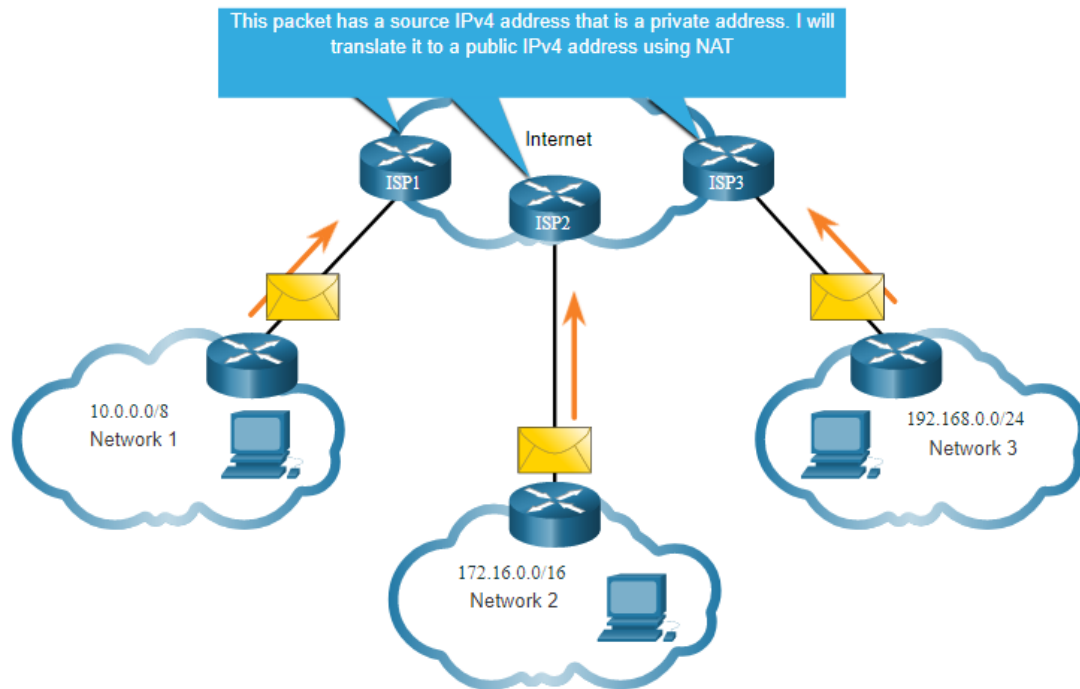
- As defined in in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.
- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
- Private IPv4 addresses are not unique and can be used internally within any network.
- However, private addresses are not globally routable.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

Types of IPv4 Addresses

Routing to the Internet

- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
- NAT is typically enabled on the edge router connecting to the internet.
- It translates the internal private address to a public global IP address.



Special Use IPv4 Addresses

Loopback addresses

- 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
- Commonly identified as only 127.0.0.1
- Used on a host to test if TCP/IP is operational.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

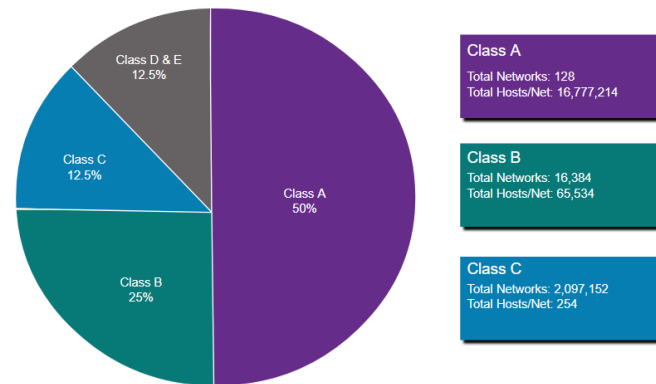
Link-Local addresses

- 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
- Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- Used by Windows DHCP clients to self-configure when no DHCP servers are available.

Legacy Classful Addressing

RFC 790 (1981) allocated IPv4 addresses in classes

- Class A (0.0.0.0/8 to 127.0.0.0/8)
 - Class B (128.0.0.0 /16 – 191.255.0.0 /16)
 - Class C (192.0.0.0 /24 – 223.255.255.0 /24)
 - Class D (224.0.0.0 to 239.0.0.0)
 - Class E (240.0.0.0 – 255.0.0.0)
-
- Classful addressing wasted many IPv4 addresses.

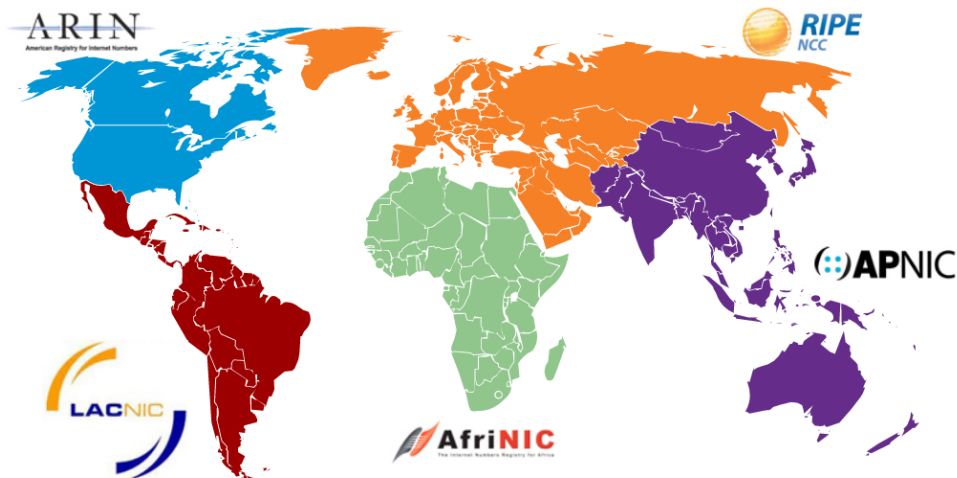


Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

Types of IPv4 Addresses

Assignment of IP Addresses

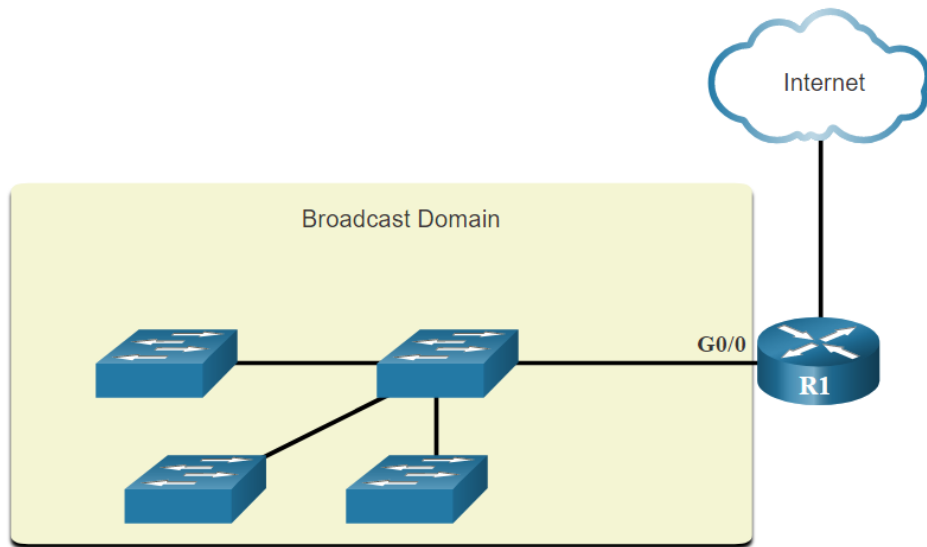
- The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).
- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.



Network Segmentation

Broadcast Domains and Segmentation

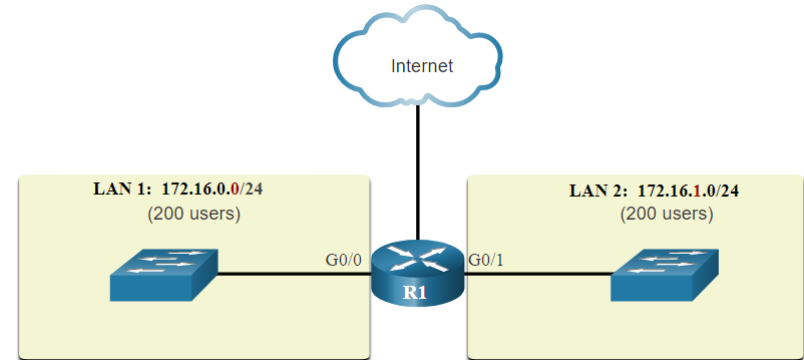
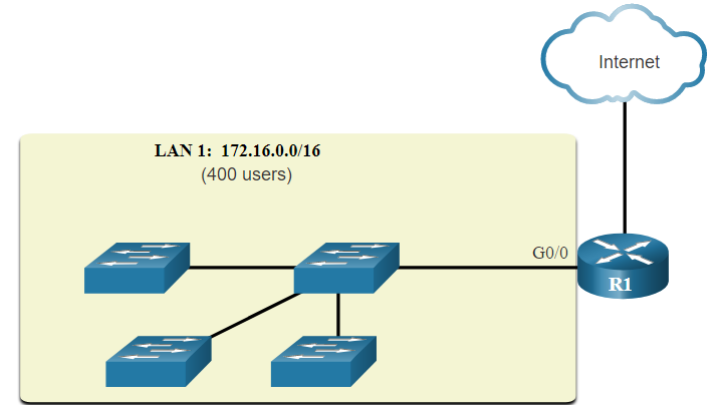
- Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
- Switches propagate broadcasts out all interfaces except the interface on which it was received.



- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.

Problems with Large Broadcast Domains

- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
- Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.
- Broadcasts are only propagated within the smaller broadcast domains.

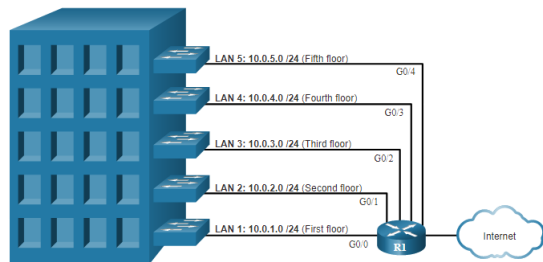


Network Segmentation

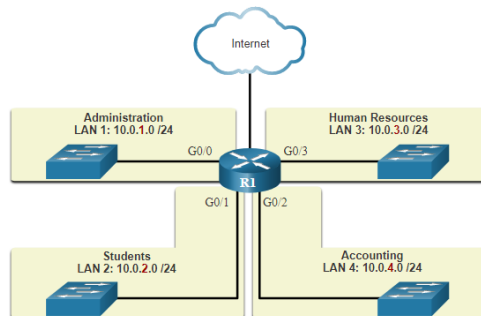
Reasons for Segmenting Networks

- Subnetting reduces overall network traffic and improves network performance.
- It can be used to implement security policies between subnets.
- Subnetting reduces the number of devices affected by abnormal broadcast traffic.
- Subnets are used for a variety of reasons including by:

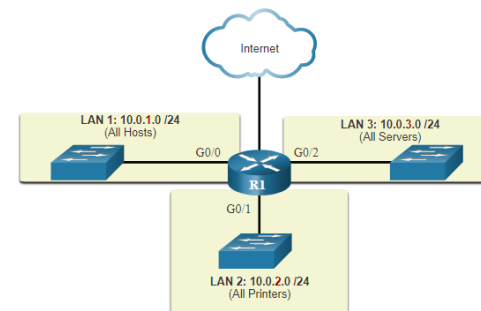
Location



Group or Function



Device Type

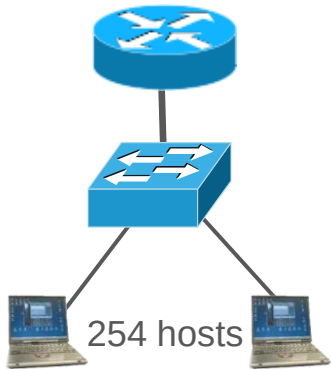


Subnet an IPv4 Network

Classful subnetting

192.168.1.0 /24

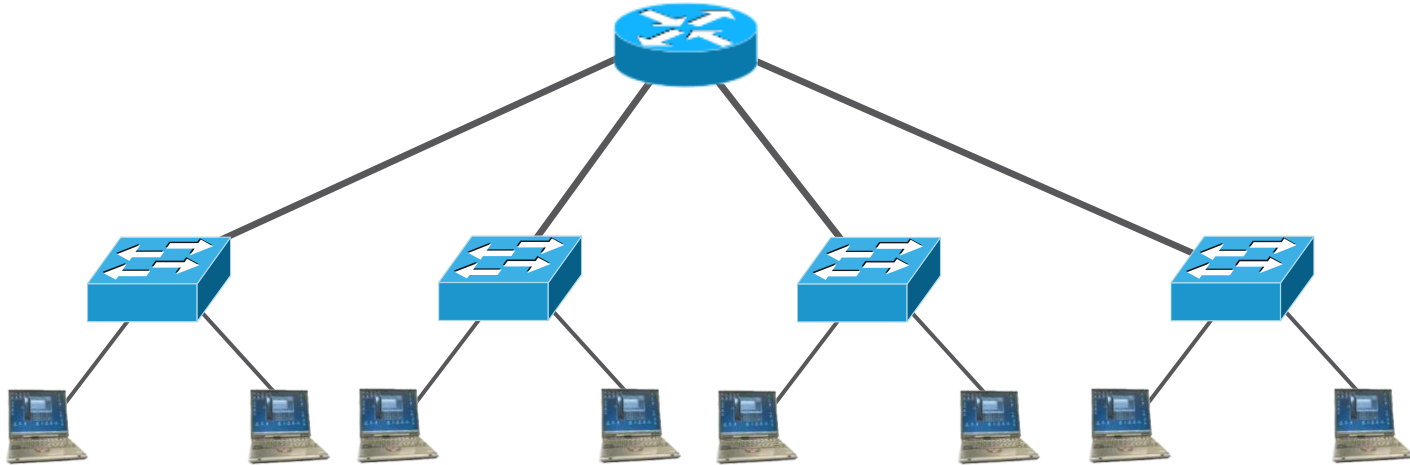
Network portion	Host portion
192.168.1	0
11000000.10101000.00000001	00000000
Nombre d'adresses de hosts possibles	$2^8 - 2 = 254$ hosts (*)



(*) all 0's not allowed and all 1s reserved for multicast

Fixed subnetting – borrowing 2 bits (2 bits de plus pour le champ subnet)

192.168.1.<2bits> = 11000000.10101000.00000001.xx

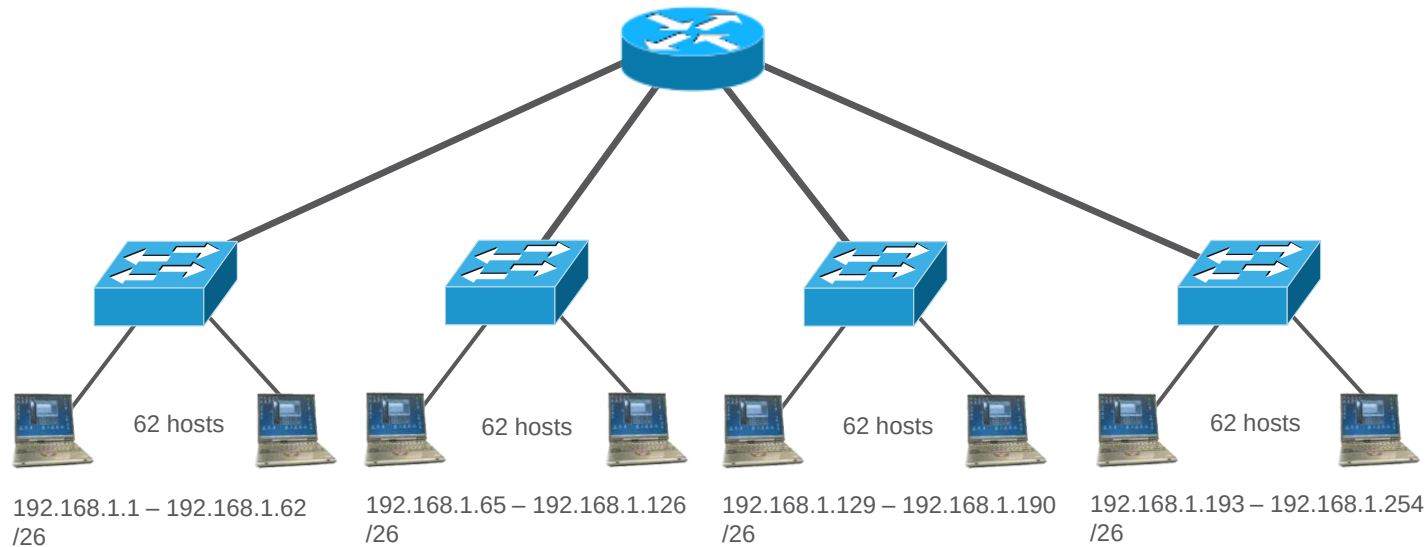


Fixed subnetting – borrowing 2 bits

Network portion	Host portion	Subnet
11000000.10101000.00000001.xx	000001 – 111110	/26
11000000.10101000.00000001.00	11000000.10101000.00000001.0000001 11000000.10101000.00000001.00111110 192.168.1.1 – 192.168.1.62	192.168.1.0 /26
11000000.10101000.00000001.01	11000000.10101000.00000001.0100001 11000000.10101000.00000001.01111110 192.168.1.65 – 192.168.1.126	192.168.1.64 /26
11000000.10101000.00000001.10	11000000.10101000.00000001.1000001 11000000.10101000.00000001.10111110 192.168.1.129 – 192.168.1.190	192.168.1.128 /26
11000000.10101000.00000001.11	11000000.10101000.00000001.1100001 11000000.10101000.00000001.11111110 192.168.1.193 – 192.168.1.254	192.168.1.192 /26

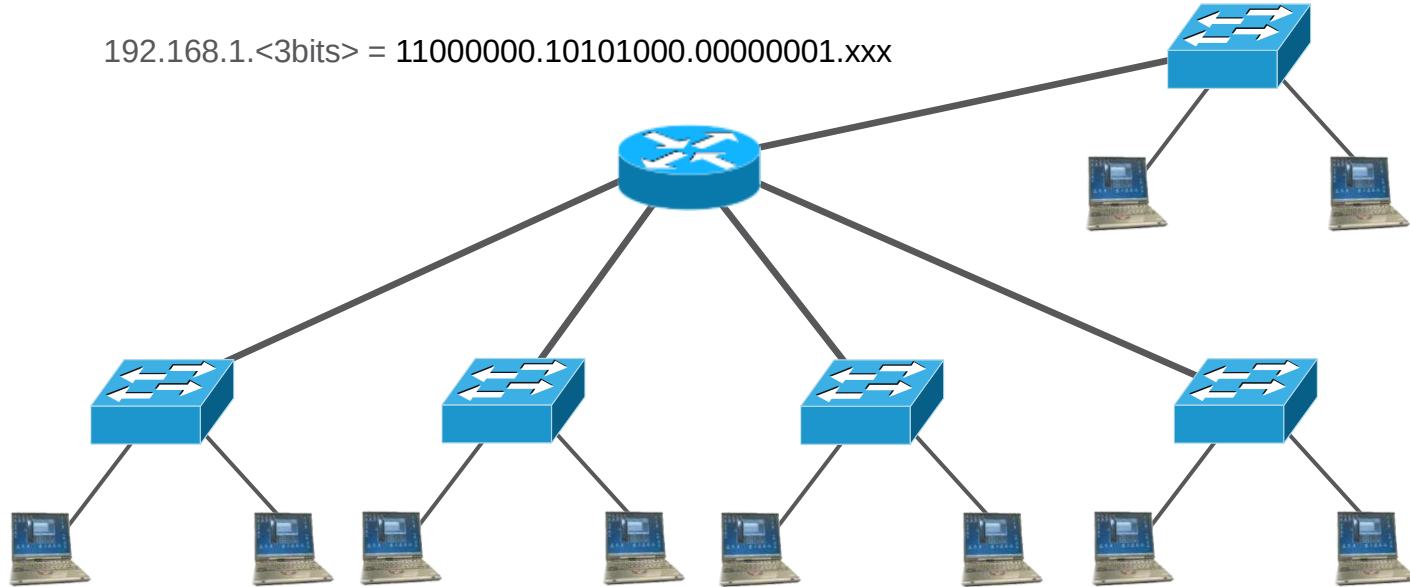
(*) all 0's not allowed and all 1s reserved for multicast

Fixed subnetting – borrowing 2 bits



Variable subnetting – borrowing 1 more bit

192.168.1.<3bits> = 11000000.10101000.00000001.xxx

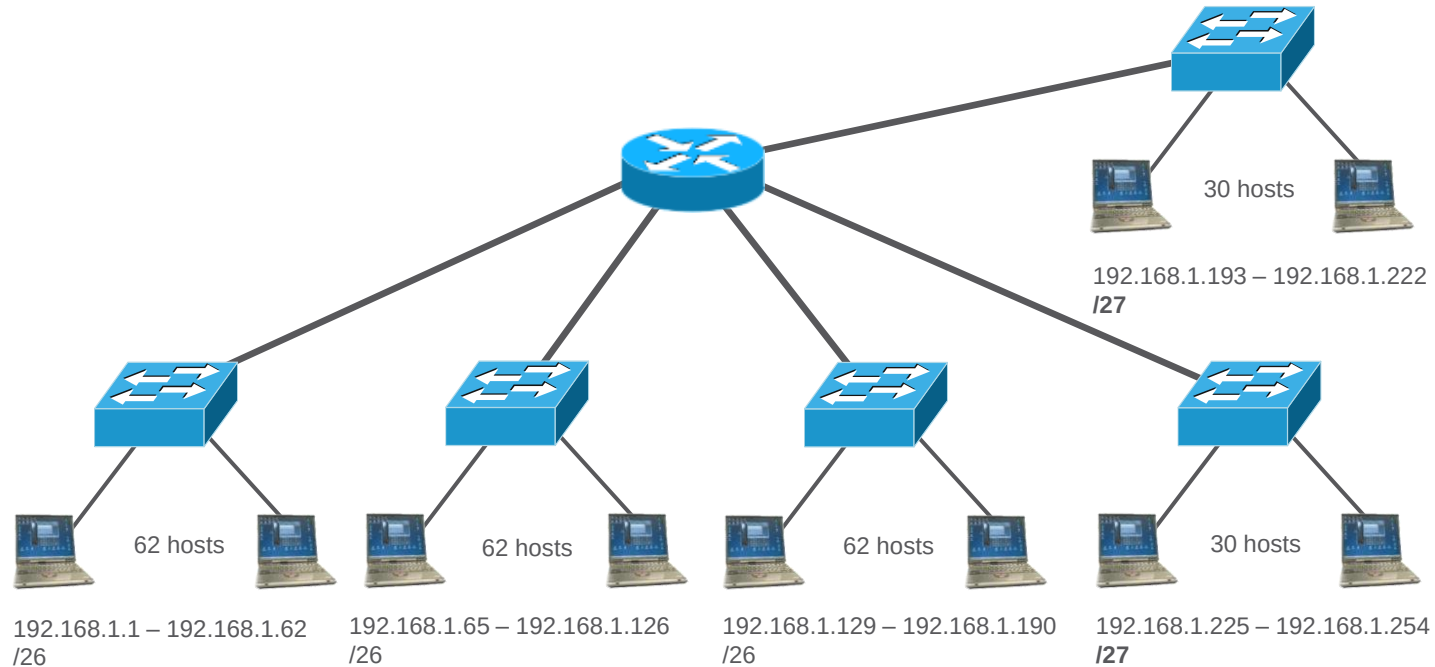


Variable subnetting – borrowing 1 more bit

(*) all 0's not allowed and all 1s reserved for multicast

Network portion	Host portion	Subnet
11000000.10101000.00000001.xx	000001 – 111110 $2^6\text{bits}-2 = 62$ hosts (*)	/26
11000000.10101000.00000001.xxx	00001 – 11110 $2^5\text{bits}-2 = 30$ hosts (*)	/27
11000000.10101000.00000001.00	11000000.10101000.00000001.00000001 11000000.10101000.00000001.00111110 192.168.1.1 – 192.168.1.62	192.168.1.0 /26
11000000.10101000.00000001.01	11000000.10101000.00000001.01000001 11000000.10101000.00000001.01111110 192.168.1.65 – 192.168.1.126	192.168.1.64 /26
11000000.10101000.00000001.10	11000000.10101000.00000001.10000001 11000000.10101000.00000001.10111110 192.168.1.129 – 192.168.1.190	192.168.1.128 /26
11000000.10101000.00000001.11x 11000000.10101000.00000001.110 11000000.10101000.00000001.111	11000000.10101000.00000001.11000001 11000000.10101000.00000001.11011110 192.168.1.193 – 192.168.1.222 11000000.10101000.00000001.11100001 11000000.10101000.00000001.11111110 192.168.1.225 – 192.168.1.254	192.168.1.192 /27 192.168.1.224 /27

Variable subnetting – borrowing 1 more bit

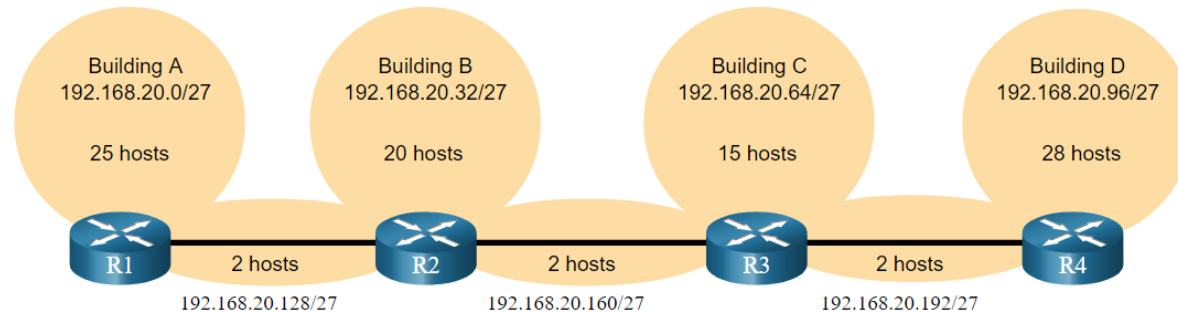


VLSM

VLSM IPv4 Address Conservation

Given the topology, 7 subnets are required (i.e, four LANs and three WAN links) and the largest number of host is in Building D with 28 hosts.

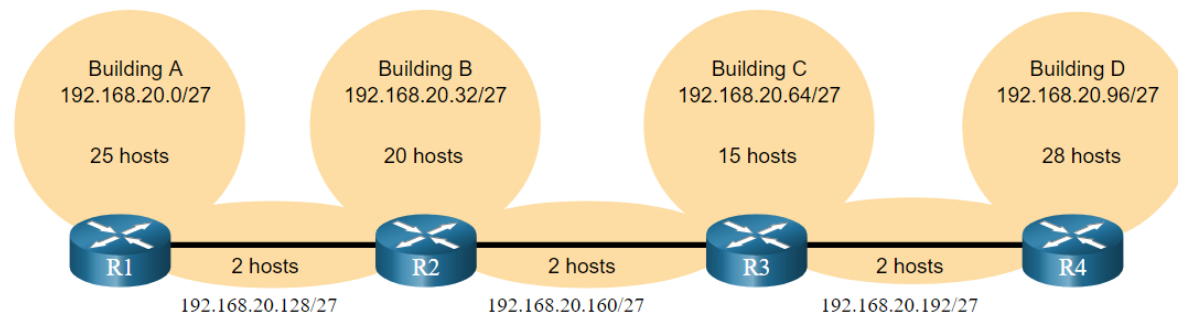
- A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology.



IPv4 Address Conservation (Cont.)

However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses.

Host portion
 $2^5 - 2 = 30$ host IP addresses per subnet
 $30 - 2 = 28$
Each WAN subnet wastes 28 addresses
 $28 \times 3 = 84$
84 addresses are unused



- Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
- VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet.

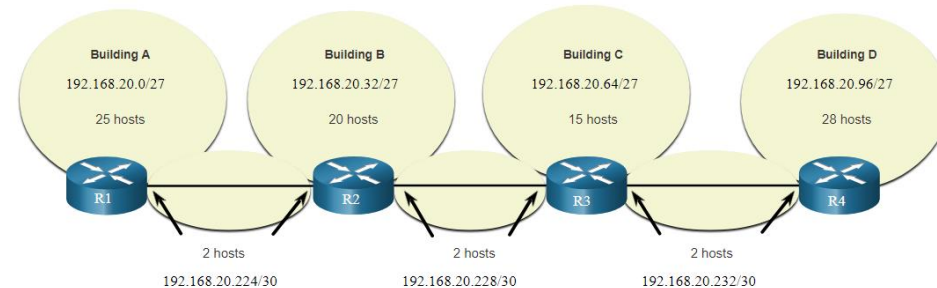
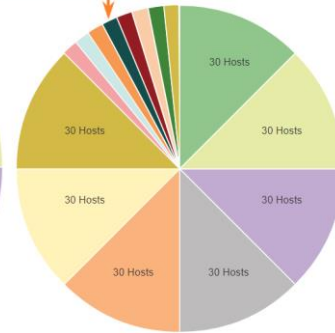
- The left side displays the traditional subnetting scheme (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to subnet a subnet and divided the last subnet into eight /30 subnets.
- When using VLSM, always begin by satisfying the host requirements of the largest subnet and continue subnetting until the host requirements of the smallest subnet are satisfied.
- The resulting topology with VLSM applied.

Traditional Subnetting Creates Equal Sized Subnets



Subnets of Varying Sizes

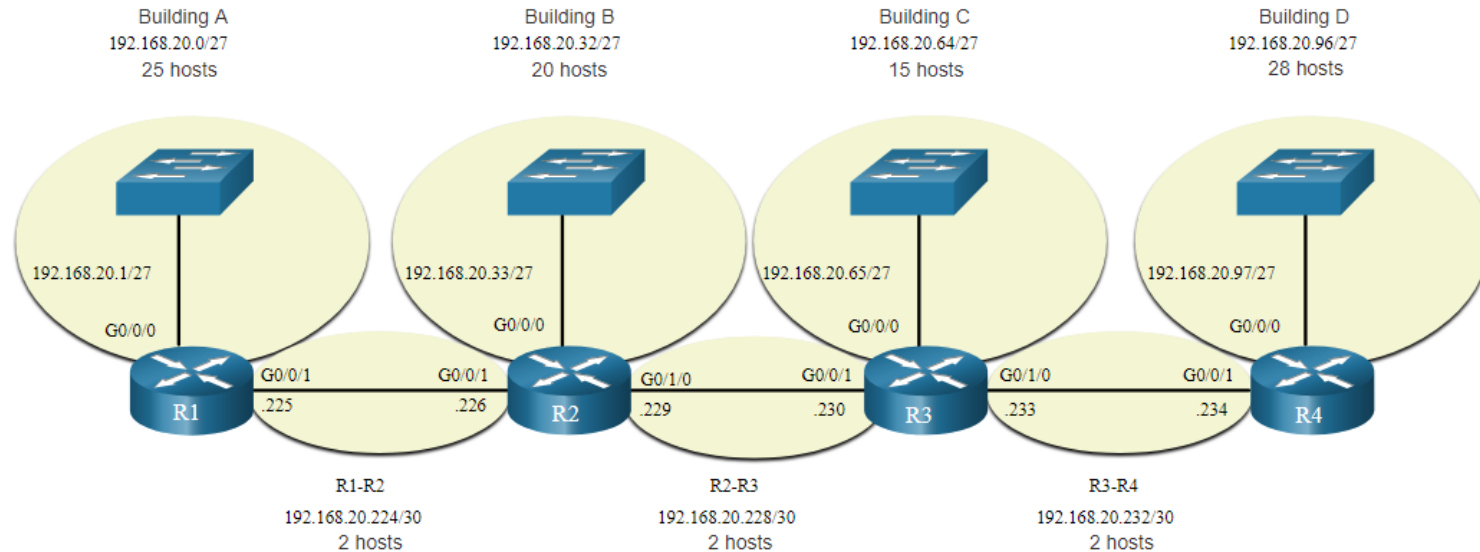
One subnet was further divided to create 8 smaller subnets of 2 hosts each.



VLSM

VLSM Topology Address Assignment

- Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram.



IPv4 Issues

IPv4 Issues

Need for IPv6

- IPv4 is running out of addresses. IPv6 is the successor to IPv4. IPv6 has a much larger 128-bit address space.
- The development of IPv6 also included fixes for IPv4 limitations and other enhancements.
- With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT, the time has come to begin the transition to IPv6.



IPv4 and IPv6 Coexistence

Both IPv4 and IPv6 will coexist in the near future and the transition will take several years.

The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6. These migration techniques can be divided into three categories:

- **Dual stack** -The devices run both IPv4 and IPv6 protocol stacks simultaneously.
- **Tunneling** – A method of transporting an IPv6 packet over an IPv4 network. The IPv6 packet is encapsulated inside an IPv4 packet.
- **Translation** - Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4.

Note: Tunneling and translation are for transitioning to native IPv6 and should only be used where needed. The goal should be native IPv6 communications from source to destination.

IPv6 Address Representation

IPv6 Address Representation

IPv6 Addressing Formats

- IPv6 addresses are 128 bits in length and written in hexadecimal.
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.
- The preferred format for writing an IPv6 address is x:x:x:x:x:x:x:x, with each “x” consisting of four hexadecimal values.
- In IPv6, a hextet is the unofficial term used to refer to a segment of 16 bits, or four hexadecimal values.
- Examples of IPv6 addresses in the preferred format:

2001:0db8:0000:1111:0000:0000:0000:0200

2001:0db8:0000:00a3:abcd:0000:0000:1234

Rule 1 – Omit Leading Zero

The first rule to help reduce the notation of IPv6 addresses is to omit any leading 0s (zeros).

Examples:

- 01ab can be represented as 1ab
- 09f0 can be represented as 9f0
- 0a00 can be represented as a00
- 00ab can be represented as ab

Note: This rule only applies to leading 0s, NOT to trailing 0s, otherwise the address would be ambiguous.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
No leading zeros	2001 : db8 : 0 : 1111 : 0 : 0 : 0 : 200

IPv6 Address Representation

Rule 2 – Double Colon

A double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all zeros.

Example:

- 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1

Note: The double colon (::) can only be used once within an address, otherwise there would be more than one possible resulting address.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
Compressed	2001:db8:0:1111::200

IPv6 Address Types

IPv6 Address Types

Unicast, Multicast, Anycast

There are three broad categories of IPv6 addresses:

- **Unicast** – Unicast uniquely identifies an interface on an IPv6-enabled device.
- **Multicast** – Multicast is used to send a single IPv6 packet to multiple destinations.
- **Anycast** – This is any IPv6 unicast address that can be assigned to multiple devices. A packet sent to an anycast address is routed to the nearest device having that address.

Note: Unlike IPv4, IPv6 does not have a broadcast address. However, there is an IPv6 all-nodes multicast address that essentially gives the same result.

IPv6 Address Types

IPv6 Prefix Length

Prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.

The IPv6 prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64.



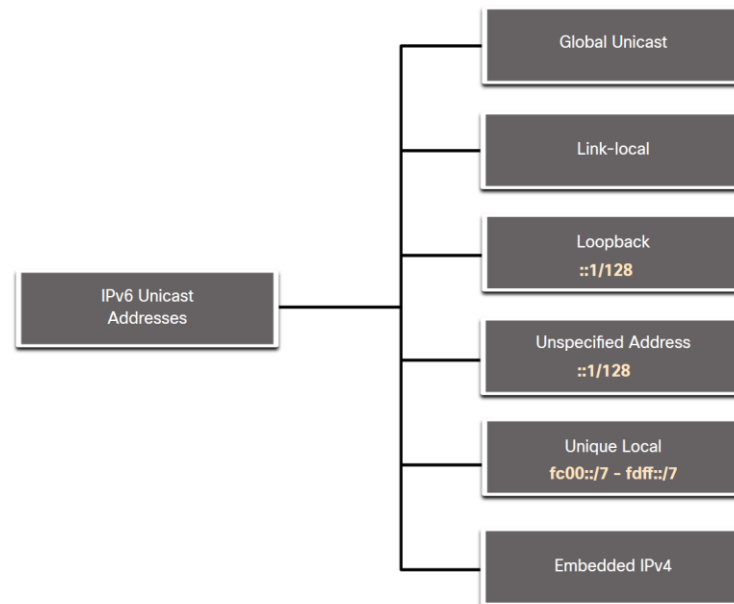
Note: It is strongly recommended to use a 64-bit Interface ID for most networks. This is because stateless address autoconfiguration (SLAAC) uses 64 bits for the Interface ID. It also makes subnetting easier to create and manage.

IPv6 Address Types

Types of IPv6 Unicast Addresses

Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:

- **Global Unicast Address (GUA)** – This is similar to a public IPv4 address. These are globally unique, internet-routable addresses.
- **Link-local Address (LLA)** - Required for every IPv6-enabled device and used to communicate with other devices on the same local link. LLAs are not routable and are confined to a single link.



A Note About the Unique Local Address

The IPv6 unique local addresses (range fc00::/7 to fdff::/7) have some similarity to RFC 1918 private addresses for IPv4, but there are significant differences:

- Unique local addresses are used for local addressing within a site or between a limited number of sites.
- Unique local addresses can be used for devices that will never need to access another network.
- Unique local addresses are not globally routed or translated to a global IPv6 address.

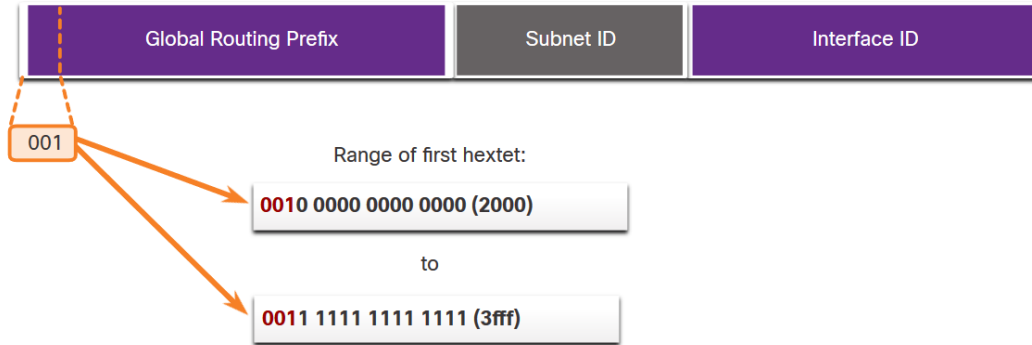
Note: Many sites use the private nature of RFC 1918 addresses to attempt to secure or hide their network from potential security risks. This was never the intended use of ULAs.

IPv6 Address Types

IPv6 GUA

IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet.

- Currently, only GUAs with the first three bits of 001 or 2000::/3 are being assigned.
- Currently available GUAs begins with a decimal 2 or a 3 (This is only 1/8th of the total available IPv6 address space).



IPv6 GUA Structure

Global Routing Prefix:

- The global routing prefix is the prefix, or network, portion of the address that is assigned by the provider, such as an ISP, to a customer or site. The global routing prefix will vary depending on ISP policies.

Subnet ID:

- The Subnet ID field is the area between the Global Routing Prefix and the Interface ID. The Subnet ID is used by an organization to identify subnets within its site.

Interface ID:

- The IPv6 interface ID is equivalent to the host portion of an IPv4 address. It is strongly recommended that in most cases /64 subnets should be used, which creates a 64-bit interface ID.

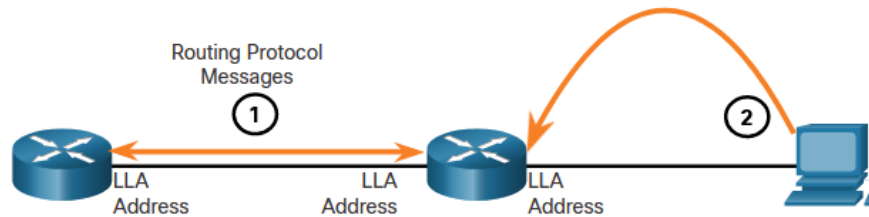
Note: IPv6 allows the all-0s and all-1s host addresses can be assigned to a device. The all-0s address is reserved as a Subnet-Router anycast address, and should be assigned only to routers.

IPv6 Address Types

IPv6 LLA

An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet).

- Packets with a source or destination LLA cannot be routed.
- Every IPv6-enabled network interface must have an LLA.
- If an LLA is not configured manually on an interface, the device will automatically create one.
- IPv6 LLAs are in the fe80::/10 range.



1. Routers use the LLA of neighbor routers to send routing updates.
2. Hosts use the LLA of a local router as the default-gateway.

GUA Addressing Structure

GUA Addressing Structure

IPv6 GUA Structure

The Site Prefix or Global Routing Prefix is the first 3 hextets or 48-bits of the address. It is assigned by the service provider.

The Site Topology or Subnet ID Is the 4th hextet of the address.

The Interface ID is the last 4 hextets or 64-bits of the address. It can be manually or dynamically assigned using the EUI-64 command. (Extended Unique Identifier)

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F/64



The diagram shows the IPv6 address 2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F/64. Red brackets are used to group the address into three parts: the first three hextets (2001:0DB8:0001) are bracketed and labeled 'Global Routing Prefix'; the fourth hextet (5270) is bracketed and labeled 'Subnet ID'; and the last four hextets (0127:00AB:CAFE:0E1F) are bracketed and labeled 'Interface ID'.

Global Routing Prefix Subnet ID Interface ID

GUA Addressing Structure

IPv6 GUA Structure

First 3 bits are fixed at 001 or 200::/12 (IANA Global Routing Number)

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



IANA

Bits 16-24 identifies the Regional Registry:

- AfriNIC, APNIC, LACNIC, RIPE NCC and ARIN

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



Registry

2001:0000::/23 – IANA

2001:0200::/23 – APNIC (Asia/Pacific Region)

2001:0400::/23 – ARIN (North America Region)

2001:0600::/23 – RIPE (Europe, Middle East and Central Asia)

GUA Addressing Structure

IPv6 GUA Structure

Remaining 8-bits up to /32 identifies the ISP.

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



ISP

The 3rd hextet represents the Site/Customer Identifier.

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



Site

*The 4th hextet represent the Site Topology/Subnet ID.

- Allows 65,536 subnets with 18,446,744,073,709,551,616 (18 quintillion) for each subnet.

- Not part of the host address field.

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



Subnet

GUA Addressing Structure

IPv6 GUA Structure

2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64



The diagram shows the IPv6 address 2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64. A red bracket is drawn under the last four hexadecets (0127:00AB:CAFE:0E1F), with a vertical line pointing down to the text 'Interface ID'.

Interface ID

The Interface ID are the remaining 64-bits of the address.

Can be manually configured or dynamically by using the EUI-64 (Extended Unique Identifier).

The EUI-64 command uses the device 48-bits MAC Address and convert it into 64-bits by adding FF:FE in the middle of the address.

The first (network) and last (broadcast) address may be assigned to an interface. An interface may contain more than one IPv6 address.

There are no broadcast addresses, multicast is used instead.

GUA and LLA Static Configuration

GUA and LLA Static Configuration

Static GUA Configuration on a Router

Most IPv6 configuration and verification commands in the Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of **ipv6** in place of **ip** within the commands.

- The command to configure an IPv6 GUA on an interface is: **ipv6 address *ipv6-address/prefix-length***.
- The example shows commands to configure a GUA on the G0/0/0 interface on R1:

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
```

GUA and LLA Static Configuration

Static GUA Configuration on a Windows Host

- Manually configuring the IPv6 address on a host is similar to configuring an IPv4 address.
- The GUA or LLA of the router interface can be used as the default gateway. Best practice is to use the LLA.

Note: When DHCPv6 or SLAAC is used, the LLA of the router will automatically be specified as the default gateway address.

Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: 2001:db8:acad:1::10

Subnet prefix length: 64

Default gateway: 2001:db8:acad:1::1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

Static GUA Configuration of a Link-Local Unicast Address

Configuring the LLA manually lets you create an address that is recognizable and easier to remember.

- LLAs can be configured manually using the **ipv6 address** *ipv6-link-local-address* **link-local** command.
- The example shows commands to configure a LLA on the G0/0/0 interface on R1

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
```

Note: The same LLA can be configured on each link as long as it is unique on that link. Common practice is to create a different LLA on each interface of the router to make it easy to identify the router and the specific interface.

Dynamic Addressing for IPv6 GUAs

Dynamic Addressing for IPv6 GUAs

RS and RA Messages

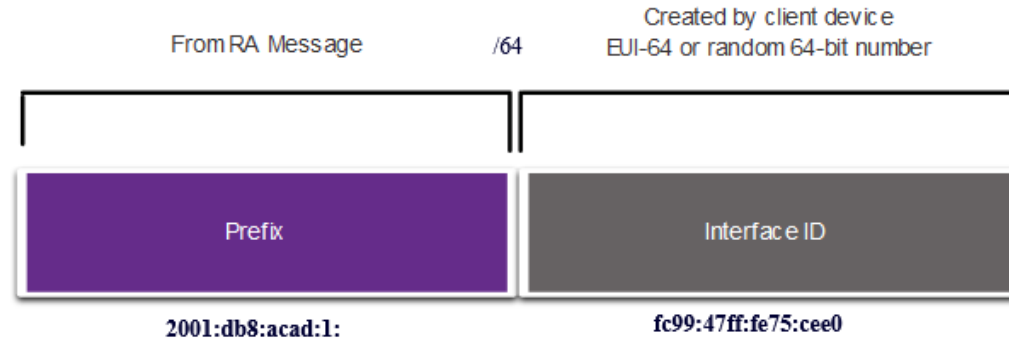
Devices obtain GUA addresses dynamically through Internet Control Message Protocol version 6 (ICMPv6) messages.

- Router Solicitation (RS) messages are sent by host devices to discover IPv6 routers
- Router Advertisement (RA) messages are sent by routers to inform hosts on how to obtain an IPv6 GUA and provide useful network information such as:
 - Network prefix and prefix length
 - Default gateway address
 - DNS addresses and domain name
- The RA can provide three methods for configuring an IPv6 GUA :
 - SLAAC
 - SLAAC with stateless DHCPv6 server
 - Stateful DHCPv6 (no SLAAC)

Dynamic Addressing for IPv6 GUAs

Method 1: SLAAC

- SLAAC allows a device to configure a GUA without the services of DHCPv6.
- Devices obtain the necessary information to configure a GUA from the ICMPv6 RA messages of the local router.
- The prefix is provided by the RA and the device uses either the EUI-64 or random generation method to create an interface ID.

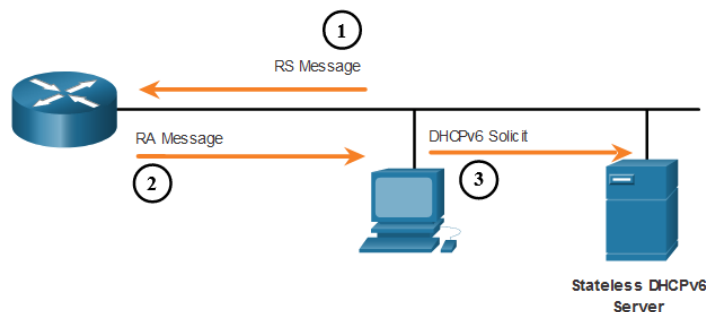


Method 2: SLAAC and Stateless DHCP

An RA can instruct a device to use both SLAAC and stateless DHCPv6.

The RA message suggests devices use the following:

- SLAAC to create its own IPv6 GUA
- The router LLA, which is the RA source IPv6 address, as the default gateway address
- A stateless DHCPv6 server to obtain other information such as a DNS server address and a domain name



Dynamic Addressing for IPv6 GUAs

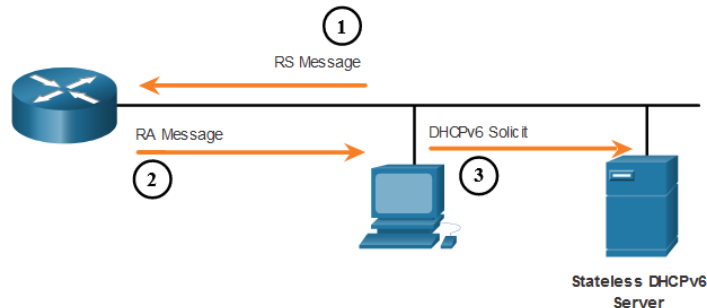
Method 3: Stateful DHCPv6

An RA can instruct a device to use stateful DHCPv6 only.

Stateful DHCPv6 is similar to DHCP for IPv4. A device can automatically receive a GUA, prefix length, and the addresses of DNS servers from a stateful DHCPv6 server.

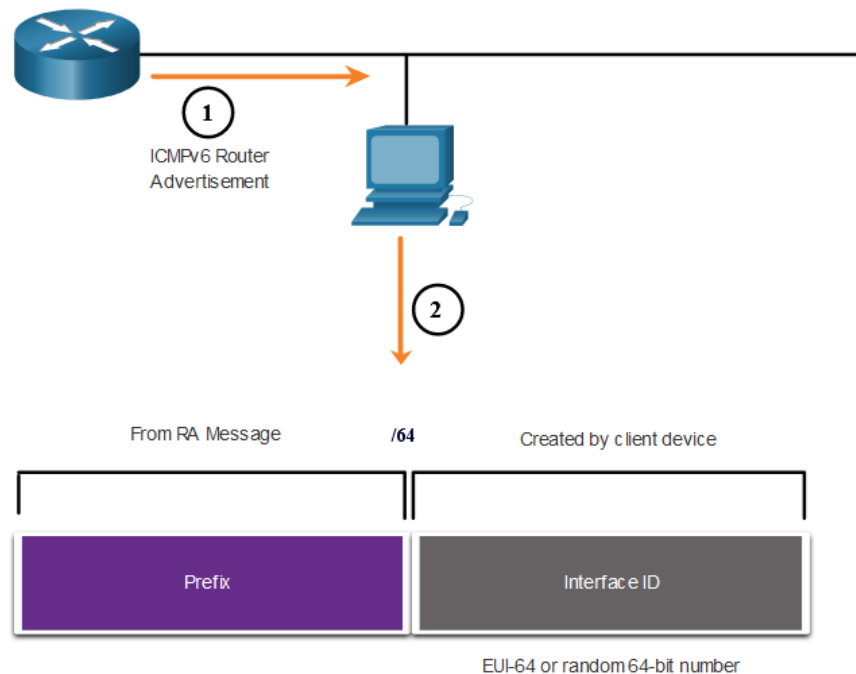
The RA message suggests devices use the following:

- The router LLA, which is the RA source IPv6 address, for the default gateway address.
- A stateful DHCPv6 server to obtain a GUA, DNS server address, domain name and other necessary information.



EUI-64 Process vs. Randomly Generated

- When the RA message is either SLAAC or SLAAC with stateless DHCPv6, the client must generate its own interface ID.
- The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number.



Dynamic Addressing for IPv6 GUAs

EUI-64 Process

The IEEE defined the Extended Unique Identifier (EUI) or modified EUI-64 process which performs the following:

- A 16 bit value of fffe (in hexadecimal) is inserted into the middle of the 48-bit Ethernet MAC address of the client.
- The 7th bit of the client MAC address is reversed from binary 0 to 1.
- Example:

48-bit MAC	fc:99:47:75:ce:e0
EUI-64 Interface ID	fe:99:47:ff:fe:75:ce:e0

Dynamic Addressing for IPv6 GUAs

Randomly Generated Interface IDs

Depending upon the operating system, a device may use a randomly generated interface ID instead of using the MAC address and the EUI-64 process.

Beginning with Windows Vista, Windows uses a randomly generated interface ID instead of one created with EUI-64.

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

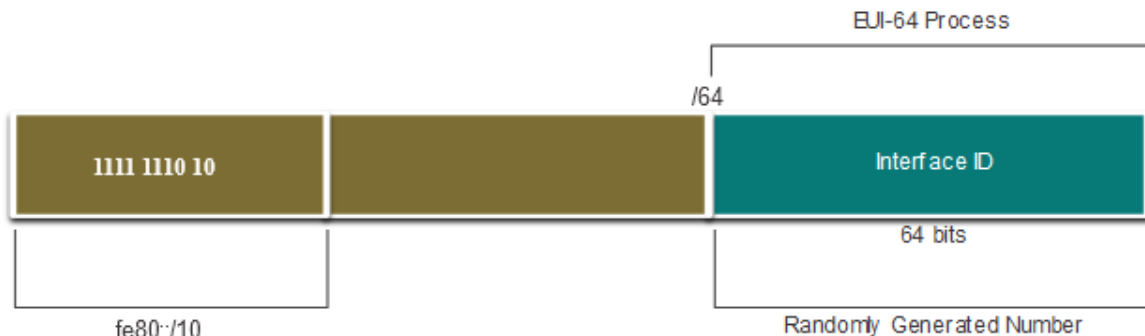
Note: To ensure the uniqueness of any IPv6 unicast address, the client may use a process known as Duplicate Address Detection (DAD). This is similar to an ARP request for its own address. If there is no reply, then the address is unique.

Dynamic Addressing for IPv6 LLAs

Dynamic Addressing for IPv6 LLAs

Dynamic LLAs

- All IPv6 interfaces must have an IPv6 LLA.
- Like IPv6 GUAs, LLAs can be configured dynamically.
- The figure shows the LLA is dynamically created using the fe80::/10 prefix and the interface ID using the EUI-64 process, or a randomly generated 64-bit number.



Dynamic Addressing for IPv6 LLAs

Dynamic LLAs on Windows

Operating systems, such as Windows, will typically use the same method for both a SLAAC-created GUA and a dynamically assigned LLA.

EUI-64 Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
Link-local IPv6 Address . . . . . : fe80::fc99:47ff:fe75:cee0
Default Gateway . . . . . : fe80::1
C:\>
```

Random 64-bit Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

Dynamic Addressing for IPv6 LLAs

Dynamic LLAs on Cisco Routers

Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.

Here is an example of a LLA dynamically configured on the G0/0/0 interface of R1:

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

Dynamic Addressing for IPv6 LLAs

Verify IPv6 Address Configuration

Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.

Here is an example of a LLA dynamically configured on the G0/0/0 interface of R1:

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

IPv6 Multicast Addresses

IPv6 Multicast Addresses

Assigned IPv6 Multicast Addresses

IPv6 multicast addresses have the prefix ff00::/8. There are two types of IPv6 multicast addresses:

- Well-Known multicast addresses
- Solicited node multicast addresses

Note: Multicast addresses can only be destination addresses and not source addresses.

Well-Known IPv6 Multicast Addresses

Well-known IPv6 multicast addresses are assigned and are reserved for predefined groups of devices.

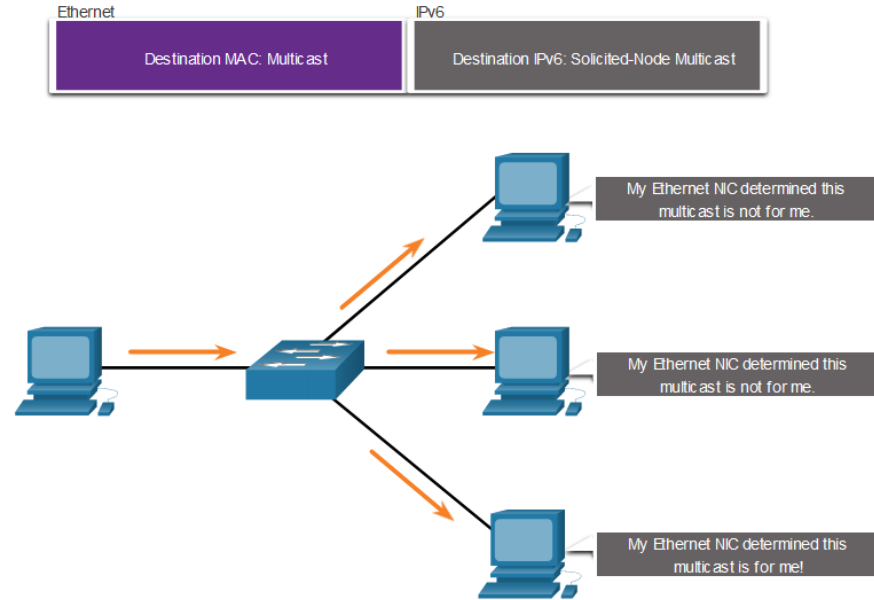
There are two common IPv6 Assigned multicast groups:

- **ff02::1 All-nodes multicast group** - This is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network.
- **ff02::2 All-routers multicast group** - This is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an IPv6 router with the **ipv6 unicast-routing** global configuration command.

IPv6 Multicast Addresses

Solicited-Node IPv6 Multicast

- A solicited-node multicast address is similar to the all-nodes multicast address.
- A solicited-node multicast address is mapped to a special Ethernet multicast address.
- The Ethernet NIC can filter the frame by examining the destination MAC address without sending it to the IPv6 process to see if the device is the intended target of the IPv6 packet.



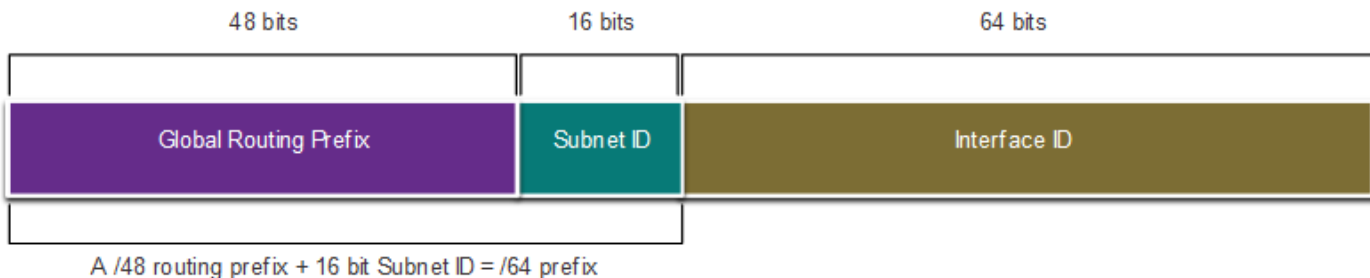
12.8 Subnet an IPv6 Network

Subnet an IPv6 Network

Subnet Using the Subnet ID

IPv6 was designed with subnetting in mind.

- A separate subnet ID field in the IPv6 GUA is used to create subnets.
- The subnet ID field is the area between the Global Routing Prefix and the interface ID.



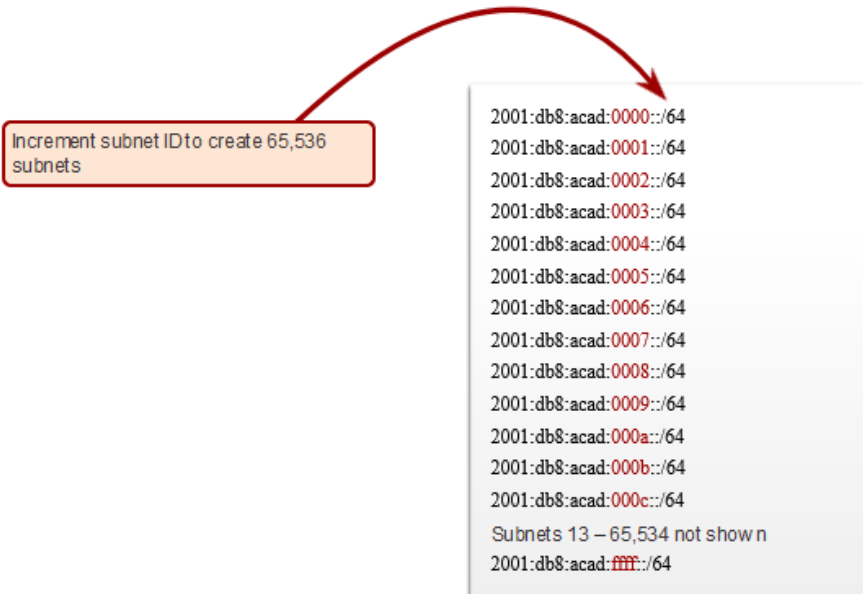
Subnet an IPv6 Network

IPv6 Subnetting Example

Given the 2001:db8:acad::/48 global routing prefix with a 16 bit subnet ID.

- Allows 65,536 /64 subnets
- The global routing prefix is the same for all subnets.
- Only the subnet ID hextet is incremented in hexadecimal for each subnet.

Increment subnet ID to create 65,536 subnets



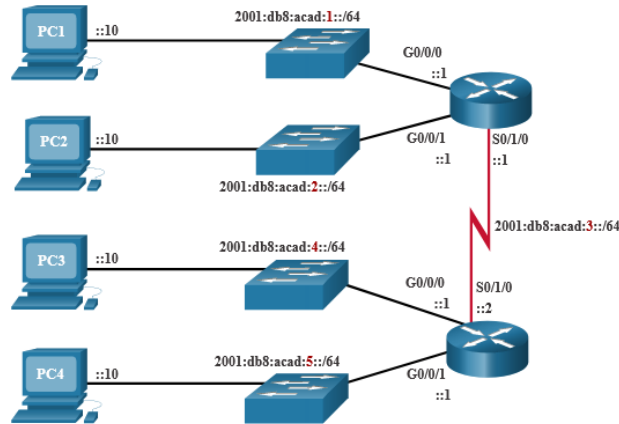
```
2001:db8:acad:0000::/64
2001:db8:acad:0001::/64
2001:db8:acad:0002::/64
2001:db8:acad:0003::/64
2001:db8:acad:0004::/64
2001:db8:acad:0005::/64
2001:db8:acad:0006::/64
2001:db8:acad:0007::/64
2001:db8:acad:0008::/64
2001:db8:acad:0009::/64
2001:db8:acad:000a::/64
2001:db8:acad:000b::/64
2001:db8:acad:000c::/64
Subnets 13 – 65,534 not shown
2001:db8:acad:fff::/64
```

Subnet an IPv6 Network

IPv6 Subnet Allocation

The example topology requires five subnets, one for each LAN as well as for the serial link between R1 and R2.

The five IPv6 subnets were allocated, with the subnet ID field 0001 through 0005. Each /64 subnet will provide more addresses than will ever be needed.



5 subnets allocated from 65,536 available subnets

Address Block `2001:0db8:acad::/48`

`2001:db8:acad:0000::/64`
`2001:db8:acad:0001::/64`
`2001:db8:acad:0002::/64`
`2001:db8:acad:0003::/64`
`2001:db8:acad:0004::/64`
`2001:db8:acad:0005::/64`
`2001:db8:acad:0006::/64`
`2001:db8:acad:0007::/64`
`2001:db8:acad:0008::/64`

`2001:db8:acad:ffff::/64`

Subnet an IPv6 Network

Router Configured with IPv6 Subnets

The example shows that each of the router interfaces on R1 has been configured to be on a different IPv6 subnet.

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```

